CT128-3-2 Implementation of Secure Systems

Introduction to Secure Network Design and Architecture

Topic & Structure

- Introduction
- Overview of Network Security Design
- Identifying Threats to Network Security
- Network Security Controls

TOPIC LEARNING OUTCOMES

At the end of this topic, you should be able to:

- Describe the goals of secure network computing:

  Confidentiality, Integrity, Availability.

- Identify the threats to network security:

  Interception, Interruption, Modification, Fabrication.

- Identify controls against network attacks:

  Physical security, Policies and procedures, and range of technical controls

Key Terms You Must Be Able To Use

- **If you have mastered this topic, you should be able to use the following terms correctly in your assignments:**
- Interception
- Interruption
- Modification
- Fabrication
- STRIDE
- Encryption
- Access Controls
- Wireless Security
- Alarms and Alert
- Honeypots

What Does "Secure" Mean?

- Protect your most valuable assets
- Placing them in a safe state
- Safe from danger / threats

What to protect?

Vulnerability

A weakness in the security system

Types of vulnerability:-

Software vulnerabilities

vulnerabilities

Hardware

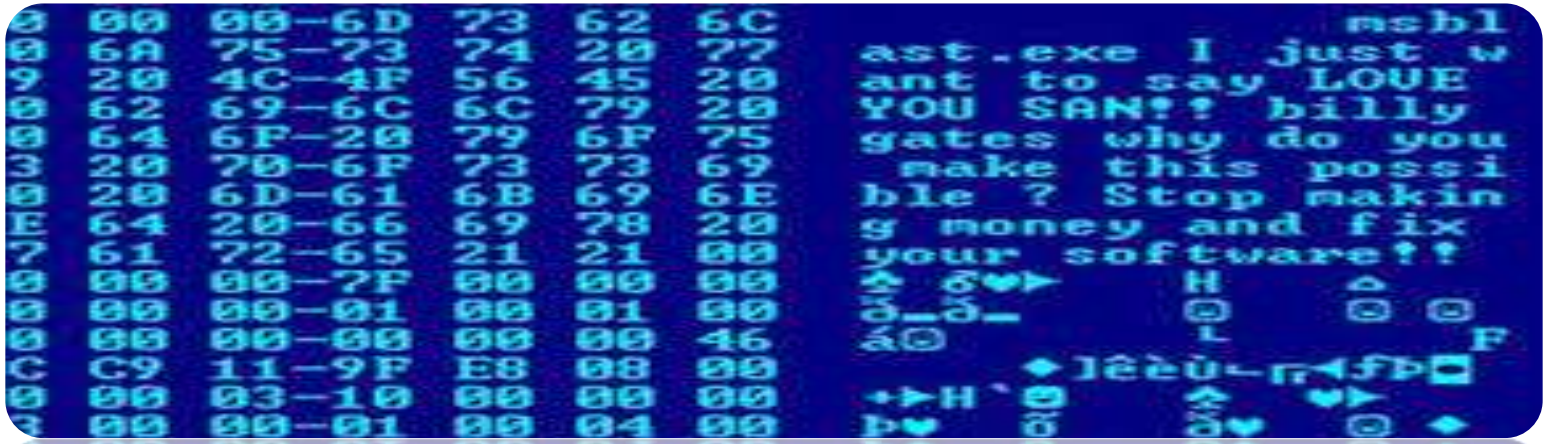
Control

- A control is an action, device, procedure, or technique that removes or reduces a vulnerability

"A threat is blocked by control of a vulnerability"

A malicious attacker must have three things

- **Method** - Knowledge of systems are widely available.
- **Opportunity**- Systems available to the public are accessible to them.
- **Motive** - A reason to want to perform this attack against this system



Security Goals

- Confidentiality
- Integrity
- Availability

Relationship between Confidentiality, Integrity, and Availability

Key Principles of Network Security

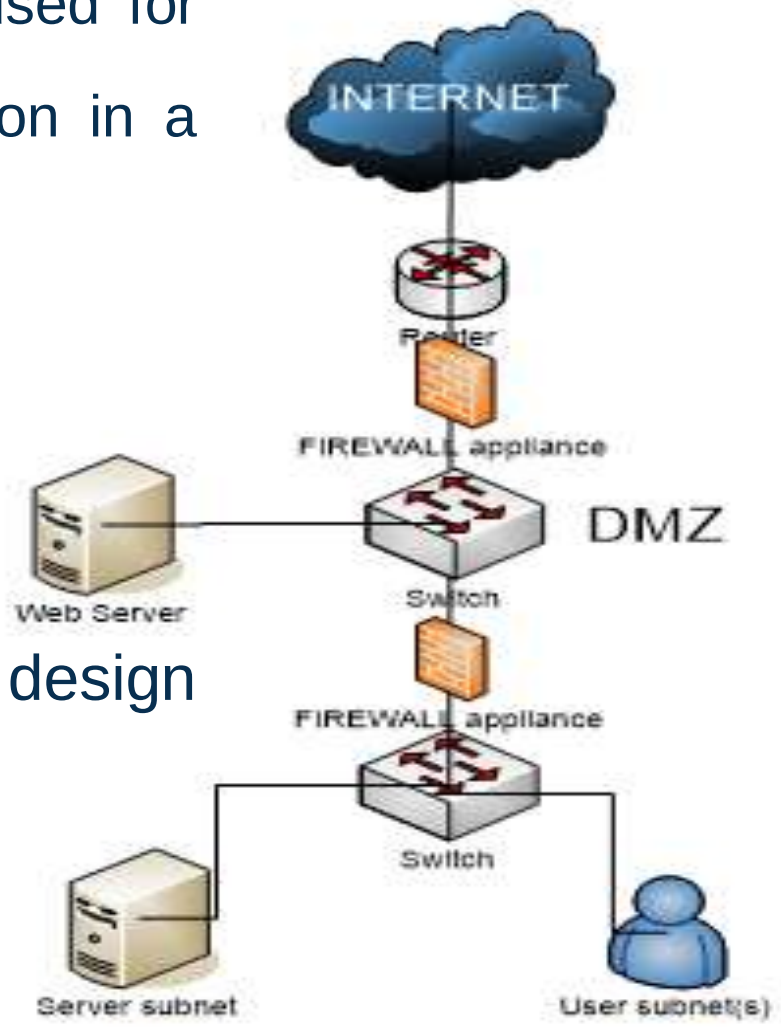
- Defense-in-Depth
  - Provides multiple layers of protection
- Least privilege / separation of duties
  - Grants the least amount of permission necessary to perform required tasks
- Minimized attack surface
  - Reduces the number of vulnerable points on the network

Security Design and Implementation

Security Design

- Ensures that a logical and carefully planned strategy is used for securing organization's assets.
- Ensures that security is applied throughout the organization in a controlled and logical manner.

- Creates policies and procedures for security.



Security Implementation

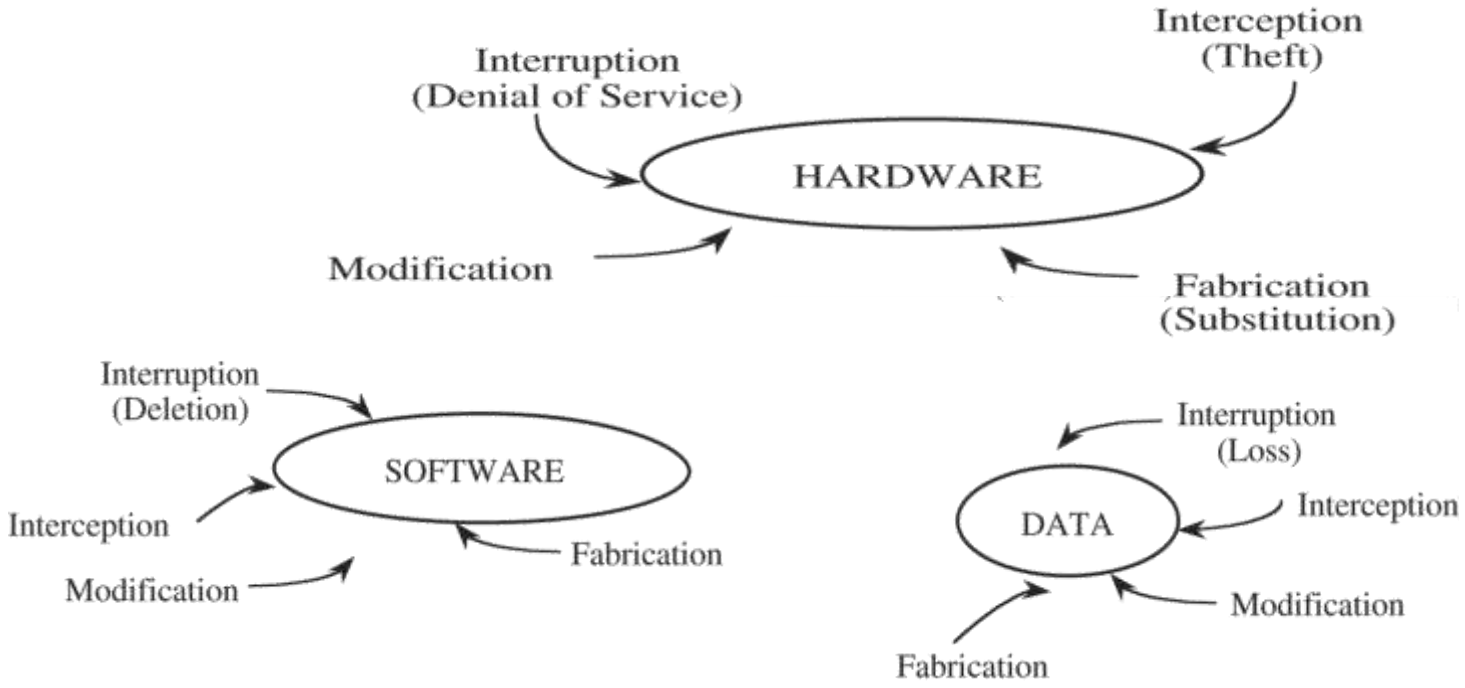
- Applies the policies and procedures created during the design to the organization's assets.

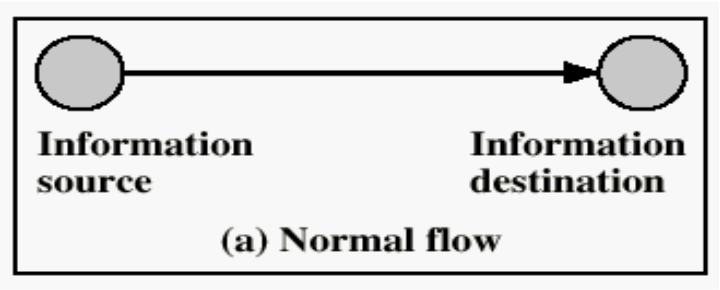
- Mechanisms to verify and monitor security controls.

Components of Network Security

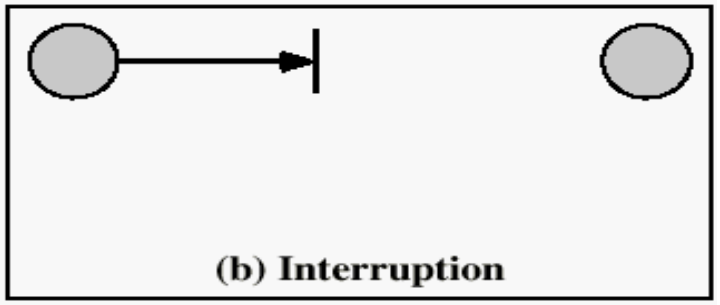
- Physical security
- Hosts
- Accounts and services
- Authentication
- Data transmission

Types of network attacks

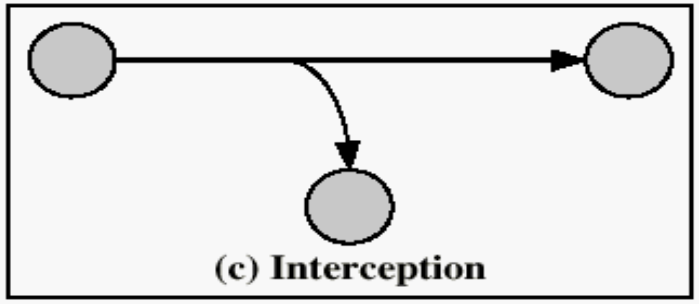




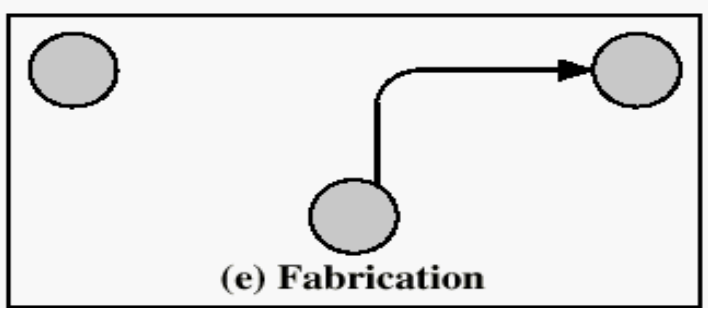
(a) Normal flow



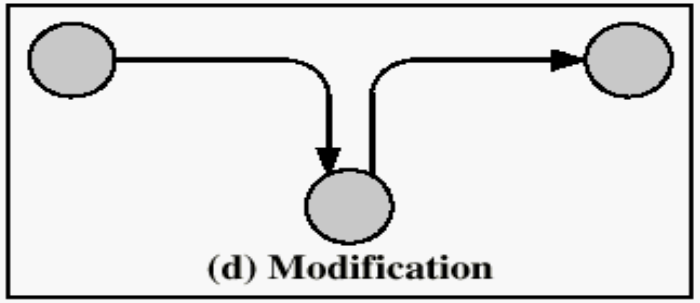
(b) Interruption



(c) Interception



(e) Fabrication

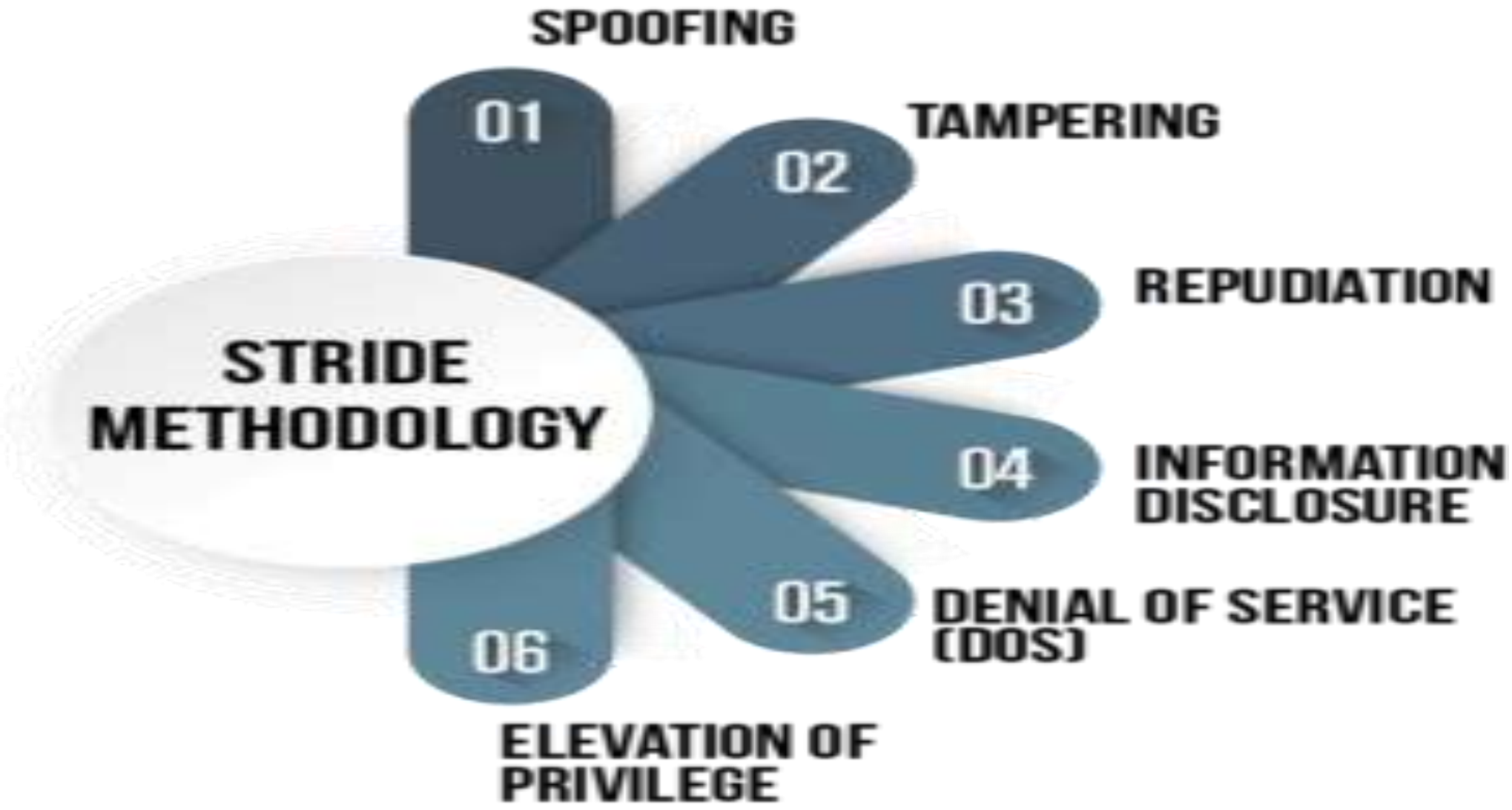


(d) Modification

Common network vulnerabilities

- Poor account passwords
- Disabled audit settings
- Using default settings

STRIDE


STRIDE METHODOLOGY
01
SPOOFING
02
TAMPERING
03
REPUDIATION
04
INFORMATION DISCLOSURE
05
DENIAL OF SERVICE (DOS)
06
ELEVATION OF PRIVILEGE

POP QUIZ !

SCAN THE QR CODE USING YOUR QRSCANNER TO ANSWER THE QUESTION

Network Security Controls

- Technical security controls protect data that is stored on the network.

Network Security Controls

- **Design and Implementation**
  - Architecture-
  - Encryption
  - Access Controls
  - Wireless Security
  - Alarms and Alert
  - Honeypots

Network Security Controls

- Design and Implementation

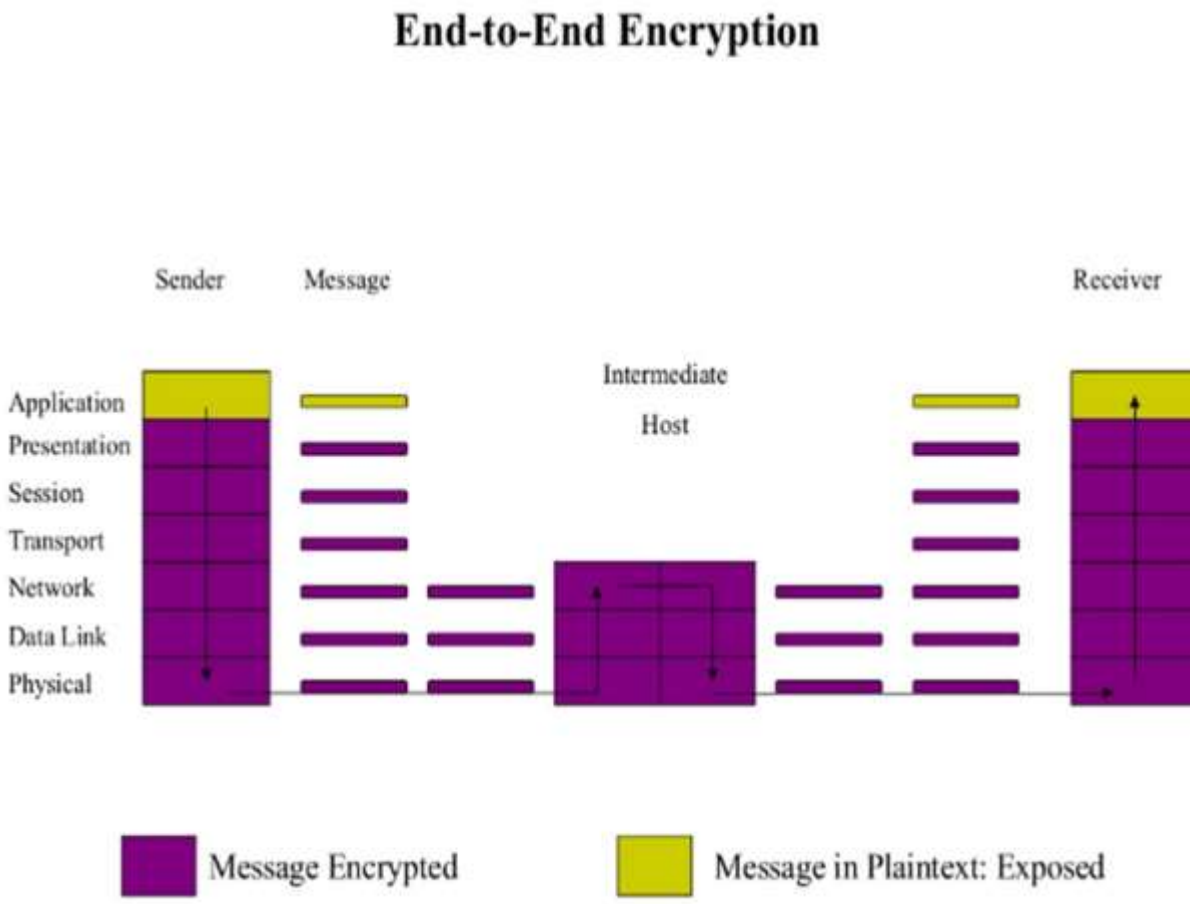
Architecture - **Network security architecture** is a framework that specifies the organizational structure, standards, policies and functional behavior of a computer **network**, including
both **security** and **network** features.

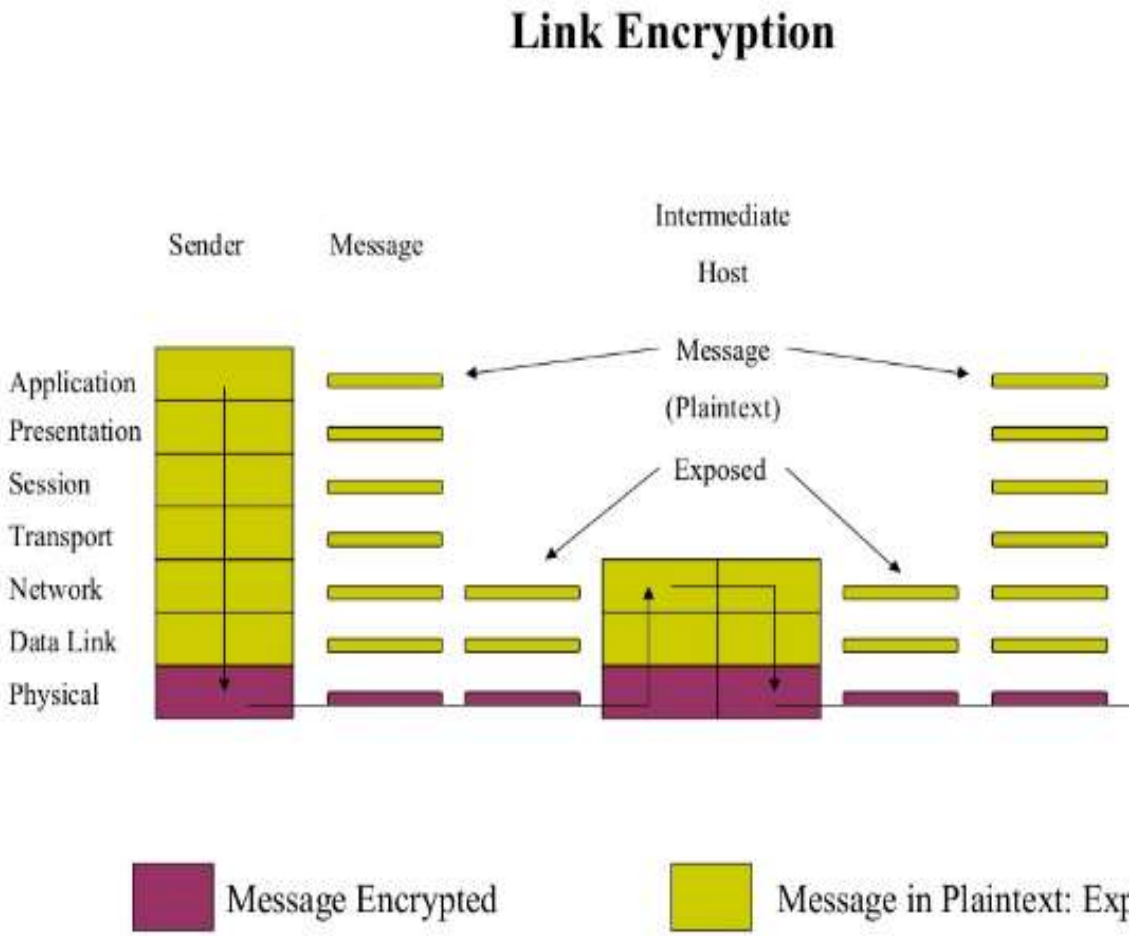
Design and Implementation

Encryption (cont..)

- **Encryption**- In network applications, encryption can be applied either between two hosts (called link encryption) or between two applications (called end-to-end encryption). Encryption keys must be delivered to the sender and receiver in a secure manner.

Design and Implementation Encryption (cont..)





Network Security Controls

- **Access controls**

Access control is a method of guaranteeing that users are who they say they are and that they have the appropriate access to company data.

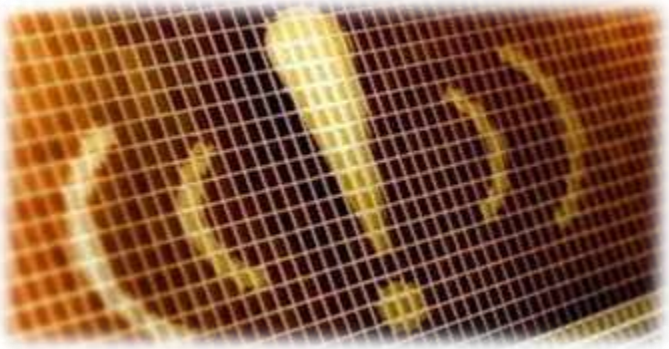
- **Wireless Security**

Wireless security is the prevention of unauthorized access or damage to computers or data using wireless networks, which include Wi-Fi networks. The most common type is Wi-Fi security, which includes Wired Equivalent Privacy and Wi-Fi Protected Access.

Network Security Controls

Alarms and Alert

– **Alert Alarm** uses a tamper-resistant GSM cellular connection, dedicated solely to your security system



Honeypots

- A **honeypot** is a controlled and safe environment for showing how attackers work and examining different types of threats.

Summary of Main Teaching Points

- Securing network communication - involves securing all digital means, hardware, software and data against four main types of network attacks; Interruption, Interception, Modification and Fabrication.
- Threat modelling is important in identifying potential threats, vulnerabilities and plan for the mitigation / control.
- STRIDE is one of the most developed and comprehensive threat modelling.
- Network security control, can be implemented in network design and its implementation which involves the architecture of network design, logical security such as encryption and access controls.

Quick Review Question

- What is mean by CIA?
- Describe the network security design?
- What are the threats to network security?
- Define the network security controls?

What To Expect Next Week

In Class

Physical Security

and

Biometrics

Preparation for Class

- **Physical Security**
  - Community roles
  - Access controls
  - Controls for protecting the secure facility
  - Fire safety, detection and response
  - Heating Ventilation and Air Conditioning-HVAC
- **Biometrics**
  - How it works
  - Fingerprints, Iris, Palm vein, Facial patterns
  - Pro & Cons
  - Application
  - Security Issues

Questions & Answer

POP QUIZ !

SCAN THE QR CODE USING YOUR QRSCANNER TO ANSWER THE QUESTION